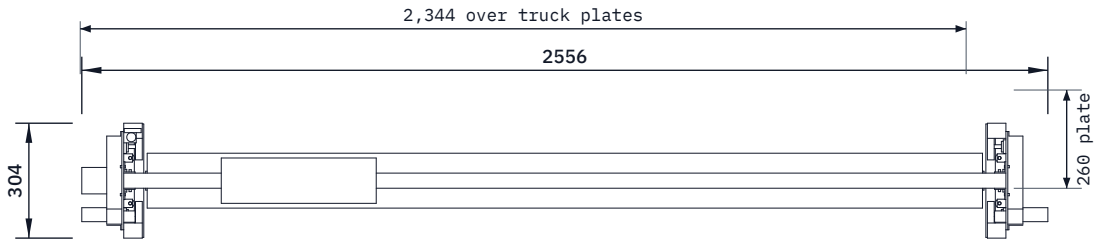


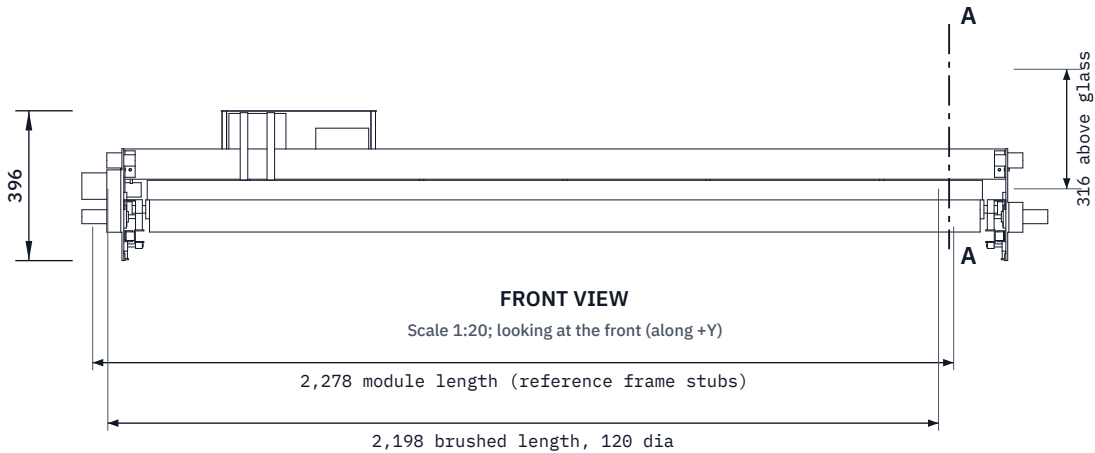
CONCEPT, NOT FOR FABRICATION

DustRunner · github.com/BoujeeEnjinia1701/dustrunner



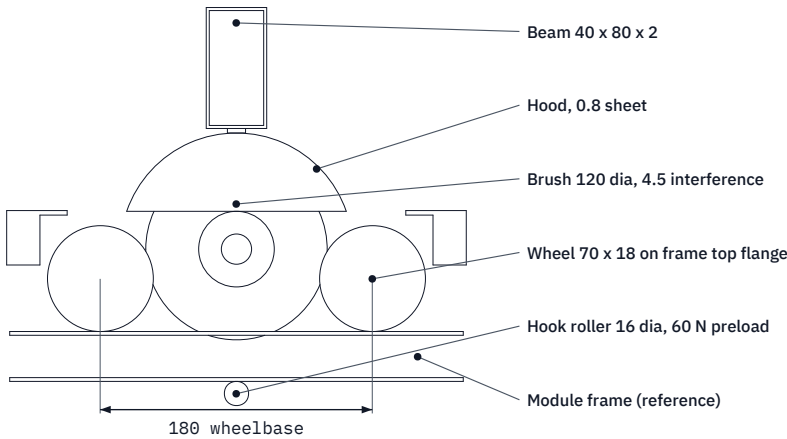
TOP VIEW

Scale 1:20; looking down (along -Z)



FRONT VIEW

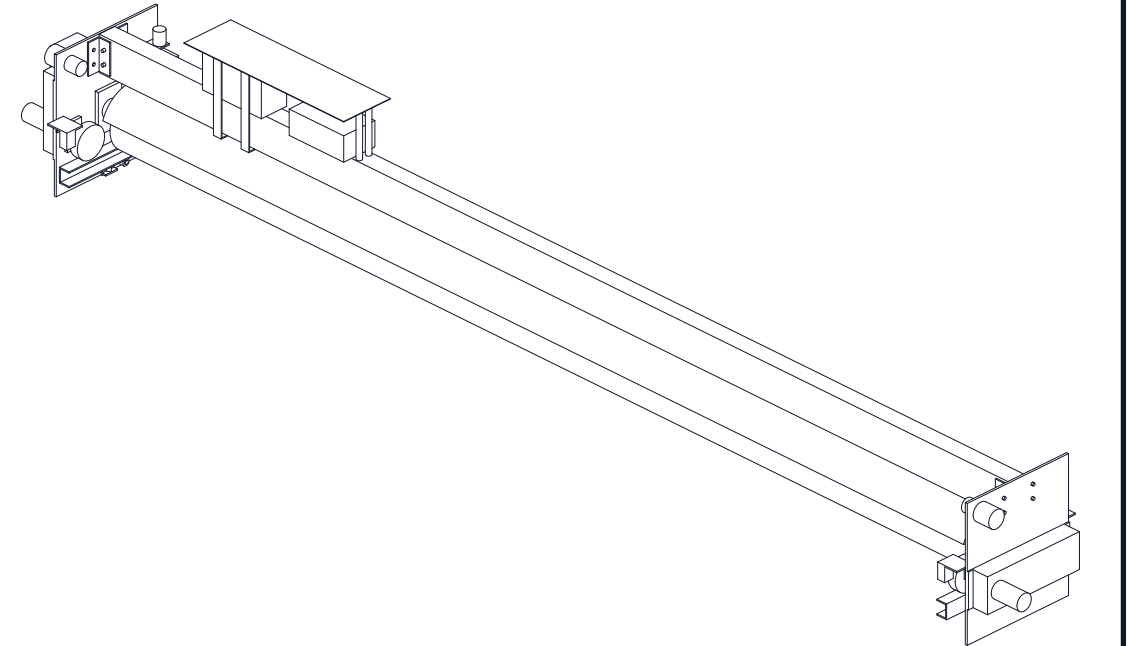
Scale 1:20; looking at the front (along +Y)



SECTION A-A

Scale 1:5; through the upper wheels, looking down the slope

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA for TRL 3 (from cad/src/model.py)	2026-09-25	AC
P2	Layout and labels tidied	2026-09-30	AC
P3	Constructable design (DRN-DDR-003): bearings, housings, clevises, sliders, cleats	2026-09-30	AC
P4	Reissued after the 2026-10-02 decisions; no geometry change (R10 mass limit 16.5 kg)	2026-10-02	AC



ISOMETRIC VIEW

Not to scale; frame stubs are the reference module; seen from the front right and above, about 30 deg elevation

MAIN DIMENSIONS AND INTERFACES (MM)

Reference module 2,278 long (1P portrait); frame 33 deep, top flange 25 wide (assumed)
Wheels 70 x 18 PU on the frame top flange, 3 to 21 from its outer face
Guide rollers 22 dia on the frame outer face; hook roller 16 dia under the bottom flange
Hook preload 60 N per truck on a sprung slider (DRN-CAL-001 C0); both wheels driven
Brush axis 55.5 above the glass: 4.5 nominal interference; core 50 x 2
Beam 40 x 80 x 2, 2,334 long, 135 to 215 above the glass
Hood 0.8 sheet, R76; nothing but the brush within 10 of the glass
Robot about 15.9 kg; about 56 N per wheel brushing (DRN-CAL-001 A3, C1)
Third-angle; sheet X up the slope, Y along the row, Z normal to the glass

DUSTRUNNER

github.com/BoujeeEnjinia1701/dustrunner

General arrangement, robot on the module frames

DRAWING NO. DRN-DWG-001		REV P4	SHEET 1 of 1	SIZE ANSI B
SCALE 1:20	UNITS mm	PROJECTION 	DATE 2026-10-02	DRAWN Amish Chadha
MATERIAL / NOTES 6063 aluminum beam, plates and hood; bought parts per bom/bom.csv. PRELIMINARY, NOT FOR FABRICATION				
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com				Generated from cad/src; do not edit by hand